INCH-POUND
MIL-DTL-82604A (OS)
21 May 2010
SUPERSEDING
MIL-C-82604 (OS)
1 March 1973

DETAIL SPECIFICATION
COMPOUND, ASPHALTIC, HIGH MELTING HOTMELT (CAVITY LINING)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a high-melting hotmelt asphaltic compound for coating the inner walls of metal cases prior to filling with explosives.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3, 4, or 5 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3, 4, or 5 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

INTERNATIONAL STANDARDIZATION AGREEMENTS

STANAG 4240 Liquid Fuel/External Fire, Munition Test Procedures

(Copies of these documents are available from NATO/MAS, Bvd Leopold 111, 1110 Brussels, BE online at https://assist.daps.dla.mil/quicksearch/, or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

Comments, suggestions, or questions on this document should be addressed to DEPARTMENT OF THE NAVY, Indian Head Division, NSWC, Code E12P, Document Control, 4123 Artisans Court, Suite 103, Indian Head, Maryland 20640-5085, OFFICIAL BUSINESS, or emailed to amanda.penn@navy.mil. Since contact information can change, you may want to verify the currency of this information using the ASSIST Online database at https://assist.daps.dla.mil.

AMSC N/A FSC 8030

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-248	Trinitrotoluene (TNT)
MIL-C-450	Coating Compound, Bituminous Solvent Type, Black (For Ammunition)
MIL-DTL-512	Aluminum Powder, Flaked, Grained, and Atomized
MIL-E-22267	Explosive Compositions, HBX Type

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-286	Propellants, Solid: Sampling, Examination and Testing
MIL-STD-1916	DOD Preferred Methods for Acceptance of Product

(Copies of these documents are available online at https://assist.daps.dla.mil/quicksearch/ or from the Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 5	Standard Test Method for Penetration of Bituminous Materials
ASTM D 36	Standard Test Method for Softening Point of Bitumen (Ring-and-Ball Apparatus)
ASTM D 92	Test Method for Flash and Fire Points by Cleveland Open Cup
ASTM D 140	Standard Practice for Sampling Bituminous Materials

(Copies of these documents are available online at http://www.astm.org or from the American Society for Testing and Materials Customer Service, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS.

3.1 General.

3.1 First article. When specified (see 6.2), a sample (see 6.3) shall be subjected to first article inspection in accordance with 4.2. A first article inspection shall be required if:

a. There has been more than 18 months since production of the latest accepted lot,
b. There has been a relocation of the production facility,
c. There has been a major change in personnel or procedures, or
d. There has been a change in the source of supplies for the materials used in the manufacture of the asphaltic compound.

3.1.2 Materials. The compound shall be an asphaltic compound. Use of a polyamide wax as an additive to adjust viscosity of the molten compound shall be permissible. The materials used in the manufacture of the asphaltic compound shall be of acceptable quality, suitable for the purpose intended, and shall be such as to produce a compound that is easily heat-liquefiable, but does not flow at temperatures slightly below the melting point. The compound shall contain no volatile solvents.

3.1.3 Conflicting requirements. Conflicting requirements arising between this specification and any specification, publication, or inspection requirement listed herein shall be referred in writing to the procuring activity or appointed agent for interpretation, clarification, and resolution.

3.1.4 Workmanship. The asphaltic compound ingredients shall be processed in a manner that will produce the highest quality material necessary to meet the requirements of this specification. The asphaltic compound shall be homogeneous and free from grit, lumps, contaminants, or other defects that could adversely affect its intended use. The asphalt compound shall be capable of storage at ambient temperature without release or separation of any component.

3.2 Product characteristics.

3.2.1 Chemical requirements. The asphaltic compound shall conform to the chemical requirements specified in table I when tested in accordance with 4.5.1.1 through 4.5.1.5.

TABLE I. Chemical requirements.

Requirement	Percent maximum
Manganese*	0.10
Lead*	0.04
Acidity, as H_2SO_4	0.01
Alkalinity, as NaOH	0.01
Insoluble matter in CCL_4	1.00
Ash	1.00

*Note: Quality conformance tests (4.5.1.1 and 4.5.1.2) for manganese and lead are not required when the ash content is less than 0.004 percent.

3.2.2 Viscosity. The viscosity of the asphaltic compound shall be a maximum of 25 Zahn seconds at 232 ± 2 ºCelsius (C) when tested in accordance with 4.5.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

3.2.3 Softening point stability. The softening point of the asphaltic compound shall be within the temperature range of 172 to 205 ºC when tested in accordance with 4.5.4.

3.2.4 Penetration. The asphaltic compound shall show the following penetration values when tested in accordance with 4.5.5:

a. 200 g/60 sec/0 ºC 5 - 35
b. 100 g/5 sec/25 ºC 5 - 35
c. 50 g/5 sec/46 ºC 80 max.

3.2.5 Flash point. The minimum flash point of the asphaltic compound shall be 300º C when tested in accordance with 4.5.6.

3.3 Performance requirements.

3.3.1 Heat resistance. A dried film of the asphaltic compound on a metal panel shall not sag or flow when heated by immersion in a silicone oil bath maintained at 172 ± 1º C for 1 hour. Test procedure shall be in accordance with 4.5.7.

3.3.2 Acid resistance. A dried film of the asphaltic compound on a metal panel shall withstand the action of picric, sulfuric, nitric, and hydrochloric acids for 6 hours without disintegration, browning, or dulling. There shall be no etching of the metal underneath the compound. Test procedure shall be in accordance with 4.5.8.

3.3.3 Water resistance. A dried film of the asphaltic compound on a metal panel shall withstand immersion in distilled water for 18 hours without blistering, whitening, and not more than a slight dulling. Test procedure shall be in accordance with 4.5.9.

3.3.4 Flexibility. A dried film of the asphaltic compound on a metal panel shall show no chipping, scaling, or cracking when tested in accordance with 4.5.10.

3.4 Reactivity requirement.

3.4.1 Reactivity. The reactivity for the asphaltic compound in contact with H-6 (MIL-E-22267) explosive and tritonal (80 ± 3 parts TNT, MIL-DTL-248, and 20 ± 3 parts aluminum, MIL-DTL-512) explosive in the vacuum stability test of 4.5.11 shall not exceed 3.0 milliliters (ml) of gas (over and above that generated by the controls).

3.5 Application - coating requirements.

a. The asphaltic compound shall be suitable for application into bombs using equipment and processing techniques that provide the interior coating requirements of applicable drawings for thermally protected bombs (see 6.4.1) in one application. The applied coating shall be self-adhering and also shall adhere to surfaces coated with cavity paint, MIL-DTL-450. Any inherent separation of components of the asphaltic compound shall not affect application or coating characteristics of the material.

b. The coated bombs (3.5a) shall be capable of being loaded with H-6 explosive to the requirements of applicable drawings for thermally protected bombs (see 6.4.1) without separation of the coating or other loss of coating effectiveness.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

c. Bombs coated with the asphaltic compound (3.5a) and loaded with H-6 explosive (3.5b) shall survive the fast cookoff test of 4.5.12. The cookoff time shall be greater than 4.0 minutes.

4. VERIFICATION.

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. The first article inspection shall consist of all the tests of this specification as described under test methods and procedures (4.5). The first article sample (4.2.1) shall meet all the requirements of this specification. The procuring activity reserves the right to accept the manufacturer's certification of testing when available, in lieu of Government testing.

4.2.1 First article sample. The first article sample shall consist of 3000 pounds of the asphaltic compound manufactured and taken from one batch of the material. The sample shall be tested by the manufacturer or forwarded to a testing facility as directed by the procuring activity (see 6.2). If the sample is to be forwarded to a testing facility, the material shall be packaged in suitable containers and shall be plainly identified by securely attached, durable labels marked with the following information:

a. Compound, asphaltic, high melting, hotmelt (cavity lining)
b. Formulation and identity of additives, if any, to the asphaltic material
c. Name and address of manufacturer
d. Location and identity of the plant that produced the samples
e. Manufacturer's identification
f. Date of manufacture
g. Submitted by (name) (date) for first article tests in accordance with the requirements of MIL-DTL-82604.

4.2.2 Fast cookoff samples. Three bombs, coated and explosive loaded to the requirements of 3.5a and 3.5b respectively, shall be forwarded to an activity designated by the procuring activity for the fast cookoff test of 4.5.12. The procuring activity shall designate the loading activity for preparing these samples.

4.3 Conformance inspection. For each inspection lot of material submitted for acceptance, conformance inspection shall consist of an examination for acceptability of quality control methods used by the manufacturer, an examination of filled containers (4.3.3) for conformance to packaging, packing, and marking requirements, and examinations and tests of samples (4.3.2) for all the requirements of 3.1.2 through 3.3.4 of this specification. The reactivity requirement of 3.4 may be a requirement for conformance inspection if at any time during a contract the procuring activity determines the material submitted under this specification to be suspect.

4.3.1 Inspection lot. An inspection lot shall consist of the same type asphaltic compound produced by one manufacturer, at one plant, from the same materials and formulation, and under essentially the same manufacturing conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

4.3.2 Sample for tests. A 10-pound sample of asphaltic compound shall be taken at random from each lot in accordance with ASTM D 140. The sample material shall be chosen at random from containers of “solidified" compound. A sample for the tests of 4.5.3 through 4.5.10, herein, shall be preconditioned for testing in accordance with 4.5.2. The sample for tests of 4.5.1 need not be preconditioned. The lot shall be unacceptable if it fails to meet all the requirements of this specification, except as noted in 4.3.

4.3.2.1 Sampling. Each container shall be labeled with complete information on the lot and batch number, date of sampling, contract number, and applicable specification.

4.3.3 Sample for examination of filled containers. A random sample of filled containers shall be selected from each lot of compound in accordance with MIL-STD-1916, Verification Level II.

4.4 Inspection procedure.

4.4.1 General. Unless otherwise specified, the values specified in section 3 apply to the results obtained from one determination for each particular test.

4.4.2 Visual examination. Conformance of the asphaltic compound to the requirement for workmanship (3.1.4) and any other requirements not covered by specific tests shall be determined by appropriate visual examination.

4.4.3 For examination, of preparation for delivery. Using the sample of filled containers selected in 4.3.3, the sample shall be adjusted to conform to MIL-STD-1916, Verification Level II. The sample unit shall be one filled and closed shipping container. Sample containers shall be examined for compliance with all requirements of this specification in regard to contents, closure, damaged or leaking container, improper container, and marking.

4.5 Test methods and procedures.

4.5.1 Chemical requirements.

4.5.1.1 Manganese. Weigh a 10- (g) sample of the compound, dissolve in 10 ml of concentrated nitric acid, and filter if necessary. Dilute to 100 ml with distilled water and transfer a 20 ml aliquot of the solution to a 250-ml beaker. Add 1 g of lead peroxide and boil. Note if manganese is present by the development of a red coloration in the solution. If the presence of manganese is indicated, filter through asbestos and titrate the clean filtrate with N/10 oxalic acid until the red coloration disappears. Calculate the percentage of manganese in the sample as follows:

$$\text{Percentage of manganese} = \frac{1.099VN}{W}$$

Where: V = ml of N/10 oxalic acid used
N = normality of oxalic acid
W = weight of sample

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

4.5.1.2 Lead. Transfer the remaining 80-ml portion of the solution obtained as described in 4.5.1.1 to a beaker and evaporate to 5 ml. Make the solution slightly alkaline with ammonium hydroxide. Acidify with acetic acid and add an excess. Heat to the boiling point and add 10 to 15 ml of a 10 percent solution of sodium or potassium dichromate. Boil the solution and allow to stand for 16 hours. Note if lead is present as indicated by a yellow precipitate. If lead is present, filter the solution through a tared Gooch crucible (or equivalent) and wash the precipitate with water and alcohol. Dry the crucible and contents at approximately 110 ºC (230 ºF) for 1 hour, cool, and weigh. Calculate the increase in weight of the crucible as lead in the sample as follows:

$$\text{Percent lead} = \frac{64.1A}{W}$$

Where: A = weight of residue, g
W = weight of original sample, g.

4.5.1.3 Acidity or alkalinity. Dissolve a 5-g sample of the compound in 10 ml of toluene, neutral to phenolphthalein. Shake the solution with 100 ml of distilled water in a 250 ml separatory funnel for at least 5 minutes. Filter and titrate water phase with N /100 H_2SO_4 or N /100 NaOH as necessary using phenolphthalein as indicator. Run a blank on an equal volume of the water used.

4.5.1.4 Insoluble matter in CCL_4.

4.5.1.4.1 Apparatus. The extraction apparatus shall consist of a heat resistant glass assay flask, a block tin condensing coil with a nickel, copper, or tin-plated metal cap that fits securely around the rim of the flask, and a glass siphon cup which is suspended under the condenser. This apparatus shall be suitable for use with an electric hotplate or steam bath.

4.5.1.4.2 Method. Transfer a sample of approximately 6 g, weighed accurately to 0.1 mg, into a tared stainless steel 200-mesh thimble. Fabricate thimble from a 15-centimeter (cm) diameter circle of wire mesh by using the lower portion of a test tube as a form for folding the thimble. This form should be of a diameter that just will fit into the glass siphon tube. Place thimble containing the sample into the siphon cup. Do not allow the thimble to extend beyond the top of the cup. Suspend the cup under the condenser by a fine wire bail. Add approximately 60 ml of carbon tetrachloride to the sample in the cups and allow the excess solvent to siphon over into the flask. Put condenser in place and circulate cold water through it. Heat the flask on a steam bath or an electric hotplate. Allow the solvent to drip from the top of the condenser at a slow to medium rate. Discontinue the extraction when the solvent drops colorless from the siphon cup. Wash down the steel mesh with a fine jet of solvent from a wash bottle. If the washings show any color, put the sample back into the siphon cup and continue the extraction until the solvent again becomes colorless. Remove the steel mesh thimble containing the residue, dry for 1 hour at approximately 100 ºC, cool, and weigh. Calculate the percent insoluble in the solvent as fo1lows:

$$\text{Percent insoluble in carbon tetrachloride} = \frac{A - B}{W} \times 100$$

Where: A = weight of steel mesh thimble plus residue after extraction, g
B = weight of steel thimble, g
C = weight of sample, g.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

4.5.1.5 Ash. Transfer a weighed portion of approximately 1 g of the sample to a tared porcelain or platinum crucible and place in a cold muffle furnace having good air circulation and capable of temperature regulation between 700 and 750 ºC. Gradually raise the temperature of the furnace to a red heat. Finish the ignition to constant weight at a temperature between 700 and 750 ºC. Cool in a desiccator and weigh. Calculate the weight of residue as percent ash in the sample.

4.5.2 Test sample preparation. The material for all of the tests required by paragraphs 3.2.2 through 3.3.4 shall be prepared at the same time and from the same melt of the test sample selected in accordance with 4.3.2 as follows.

4.5.2.1 Apparatus.

a. Sample container - Stainless steel, 1-liter (l) beaker approximately 4⅛ inches I.D. by 5¾ inches high, straight sides, no pouring lip.
b. Oven - Electric, mechanical convection type, temperature range: ambient to 250 ºC.
c. Stirring rod - 5/16-inch diameter, 12 inches long, Teflon with a steel core.

4.5.2.2 Preconditioning. Transfer a 600 to 650-g sample of the asphaltic compound selected in 4.2 to a 1-l stainless steel beaker. Heat the sample in an oven, mechanical convection type, for 3¾ hours. Set the temperature of the oven to 232 ± 2 ºC. When the oven is at the desired temperature, place the sample in the center of the oven with good air circulation on all sides. Hand-stir the sample with the Teflon stirring rod every 30 minutes to prevent lumping. At the end of the heating period, remove the sample from the oven for the Zahn viscosity test of 4.5.3. When this test is completed, return the sample and its container to the same oven and heat for an additional 30 minutes. Do not allow the sample to remain out of the oven for viscosity testing for longer than 2 minutes. At the end of the second heating period, pour the samples and coat steel panels for the tests of 4.5.4, 4.5.5, 4.5.6, 4.5.7, 4.5.8, 4.5.9, and 4.5.10.

4.5.3 Viscosity. Determine the viscosity of the asphaltic compound with a Zahn viscosimeter, No. 5 cup, at the end of the 3¾ hours of heating at 232 ± 2 ºC. Preheat the viscosimeter in the same oven and at the same time as the sample is being prepared. Remove the sample beaker containing the molten compound and the viscosimeter from the oven. Hold the viscosimeter in a vertical position and immediately dip it into the sample completely filling the cup. Lift the cup ½-inch above the top surface of the beaker and allow the asphaltic compound to drain into the beaker. With a stopwatch, record the time in seconds between when the cup breaks the surface of the sample and when the steady flow of the liquid from the orifice suddenly stops.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

4.5.4 Softening point stability. Determine softening point on two samples run in the same bath at the same time in accordance with ASTM D 36 using the double ring-holding device of ASTM D 36, except use Dow Corning Silicone Fluid DC-200 of 350 centistokes viscosity in lieu of water for the bath fluid. The thermometer shall be a total immersion, range -10 to 250 ºC, 0.5 degree divisions, 16 inches in length (Arthur H. Thomas No. 9296-L40 is satisfactory). Employ two shouldered rings and heat the samples at a rate of 5 ºC/min ± 1 ºC for the last 30 degrees before the softening point is reached. (Temperature program controller, Mode1 No, 240, Mfg. Hewlett Packard; Chromalox, 2000-watt hotplate, Mfg. Edwin L. Wiegard Co., can be used to obtain the proper heating rate.) Drill the top cover of the ring-holding device to accommodate a small glass, motor driven stirrer with a 6-inch shaft. Heat the shouldered rings to 232 ± 2 ºC before the molten compound is poured. To facilitate release of filled rings from the bottom plate of the mold, wipe the plate with silicone oil, leaving a thin film before pouring the sample. After samples have solidified, place mold on a hotplate. When mold becomes hot, but still can be handled with the bare hand, remove from the hotplate and slide rings from the bare plate to a piece of aluminum foil. Take the average of the values to determine the softening point.

4.5.5 Penetration. Determine the penetration values in accordance with ASTM D 5. Preheat ointment cups before pouring samples into them.

4.5.6 Flash point. Determine the flash point in accordance with ASTM D 92.

4.5.7 Heat resistance.

4.5.7.1 Preparation of test panel. Use a solvent-cleaned, mechanically polished, plain carbon steel panel, 6 by 4 inches, by approximately ⅛-inch thick. Preheat the panel for 5 minutes, prior to dipping in the preconditioning oven (see 4.5.2.2). Coat the panel by dipping it into the molten compound which shall be maintained at 232 ± 2 ºC. Leave the top portion of the panel uncoated. Ensure the asphalt thickness is approximately 1/16 inch on each side of the panel. Suspend the panel with the uncoated portion up and allow to cool to room temperature. Trim the excess compound from the bottom edge of the panel with a sharp knife.

4.5.7.2 Method. Suspend a panel, prepared as described in 4.5.7.1, in a bath of silicone oil, Dow Corning DC-200 (350 cs), at 172 ± 1 ºC for 1 hour so that the uncoated end of the panel is uppermost and one-half of coated portion is submerged. Remove panel and cool by immersion in cold water. Examine the film for sag or flow. A slight surface displacement at the silicone line is permissible.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

4.5.8 Acid resistance. Place a panel, prepared as described in 4.5.7.1, on a flat horizontal surface. Place drops of the following solutions at different places on the coating:

Saturated picric acid
Sulfuric acid, sp. g r. 1.30
Nitric acid, sp, gr. 1.22
Hydrochloric acid, sp. gr. 1.09

Allow the acids to remain in contact with the coating for 6 hours while the temperature is maintained at approximately 21 ºC. Cover the drops with small watch glasses in order to prevent evaporation. Remove the acids by washing with cold water and dry the film at room temperature for 1 hour. Examine the areas previously in contact with the acids and note if the compound is resistant to each of these acids as indicated by absence of disintegration, browning, or dulling of the coating. Do not consider a slight bloom around the area of a spot exposed to acid as an indication of failure to resist the acid. Remove the coating with benzene and examine the metal. Note if the coating has resisted the action of each acid as indicated by the absence of any corrosion of the metal.

4.5.9 Water resistance. Suspend a panel, prepared as described in 4.5.7.1, in a beaker of cold distilled water for 18 hours so that the uncoated end of the panel is uppermost and one-half of the coated portion is submerged. Examine the panel for blistering, whitening, softening, or other visible defects.

4.5.10 Flexibility. Clean a strip of No. 30-gage hot-rolled steel sheet, 2 by 8 inches, by washing with benzol and thoroughly drying. Coat a preheated strip on one side to a thickness of 1/16 inch by flowing on the molten compound and then cooled at room temperature for a period of 6 hours. With the coating on the outside, bend the test panel rapidly 180 degrees over a mandrel 1 inch in diameter. Ensure no macroscopic cracking of any kind is visible after the bending test.

4.5.11 Reactivity. (First article test by designated activity only unless otherwise specified.) Determine the reactivity of the asphaltic compound in contact with H-6 and tritonal explosives in accordance with the vacuum stability test procedures of 4.5.11.1 through 4.5.11.4.

4.5.11.1 Preparation of samples. Heat the compound until pourable, then deposit on glass plates a sufficient amount of the compound to provide 2.5 (N+2) g of dried film (where N equals the number of explosives involved). Air-dry the films under ambient conditions for 48 hours. After drying, remove and cut into approximately ¼-inch squares. Conduct this operation in a room that is free from acid, alkaline, or other contaminating fumes. Reduce the explosive materials to 12 mesh or finer by grinding or rasping. Keep both the compound film and the ground explosives dry by storing in a desiccator over a desiccating agent until ready for testing.

4.5.11.2 Calibration of apparatus. Calibrate the necessary number of vacuum stability test assemblies, as directed in procedure 4.1 of method 403.1.3 in MIL-STD-286.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

4.5.11.3 Testing procedure. Use 2N + 2 (where N equals the number of explosives used) tubes similar to the heating tube portion of the apparatus in procedure 4.1 of method 403.1.3 in MIL-STD-286. For controls, add 2.5 g of the dried compound to each of two tubes and 2.5 g of each explosive to additional individual tubes. Place uniform mixtures of 2.5/2.5 g of the asphaltic compound and each of the explosives specified in the test in single separate tubes. Obtain uniform distribution or mixing by carefully layering alternate portions of the dried compound with the explosive. After the addition of each portion, lightly tamp the mixture with a glass or metal rod to insure intimate contact of the materials. After all the samples have been prepared, connect the respective heating tubes with the capillary tubes as prescribed and conduct the vacuum stability test according to method 403.1.3 of MIL-STD-286 at 100 ± 0.5 ºC for 48 hours. Make all readings at room temperature with the samples removed from the bath. Correct the readings of both the controls and the test samples to standard conditions of temperature and pressure, and check for compliance with 3.4.

4.5.11.4 Calculation of reactivity. Calculate the reactivity of each of the explosive materials with the coating compound as follows:

$$\text{Reactivity in ml of gas} = X - (Y + Z)$$

Where: X = ml of gas produced by the mixture of explosive material and asphaltic compound
Y = ml of gas produced by the explosive material alone
Z = ml of gas produced by the asphaltic compound alone.

4.5.12 Fast cookoff test. The three sample bombs (4.2.2) meeting the application-coating requirements of 3.5 shall survive the fast cookoff test in accordance with STANAG 4240.

5. PACKAGING.

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DOD personnel or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service’s system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES.

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The high melting, hotmelt asphaltic compound covered by this specification is intended for use as a protective lining for fabricated metal cases in applications where charges of high explosives are poured into the case and require thermal insulation from the elevated temperature environment of a fuel fire. As this compound was developed for use in military ordnance, there is no commercial application.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, revision letter, and date of this specification
b. Quantity of asphaltic compound ordered
c. Testing activity designated to perform first article tests
d. Type and degree of contractor quality assurance program required
e. Packaging information (see 5.1 and 6.5)
f. Whether reactivity is required (see 3.4).

6.2.1 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The specific acquisition should be reviewed to ensure that only essential data are requested/provided. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Paragraph no.	Data requirement title.
4.3	Test Report
6.3	Prior approval applicability
6.5.2.2	Safety Data Sheet

6.3 First article. When a first article inspection is required, the contracting officer should provide specific guidance to offeror(s) whether the sample(s) should be a preproduction sample, a first article sample, a first production sample, a sample selected from the first production batches, or a standard production sample from the contractor's current inventory. The contracting officer should also include specific instructions in acquisition documents of the first articles. Invitation(s) for bid(s) should provide that the U.S. Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the U. S. Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior U.S. Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Definitions

6.4.1 Thermally protected bombs. Thermally protected bombs as defined in this specification are bombs coated, prior to explosive loading, with insulating materials--exterior or interior or combination of both--that provide a heat resistant barrier between the explosive charge and the elevated temperature environment of a fuel fire. The coating thickness utilizes on bombs explosive loaded to meet the bomb application—coating requirements (3.5) of this specification will be determined by the procuring activity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

6.5 Preparation for delivery.

6.5.1 Packaging and packing. The material should be packed in type I, grade C fiber drums conforming to PPP-D-723. The inside of the container should be treated with a suitable release agent which will not contaminate the contents and which will promote ready removal of the contents. Each drum should be furnished with a top heading. Drums should be palletized in conformance with MIL-STD-147.

6.5.2 Marking. Unless otherwise specified in the contractor order, marking should be in accordance with MIL-STD-290, as applicable.

6.5.2.1 Hazardous material identification. In addition to the marking required by MIL-STD-290, the fiber drums should be marked with hazardous material identification symbols in accordance with FED-STD-313.

6.5.2.2 Safety data sheet. When required by the contract or purchase order, the supplier should prepare a safety data sheet in accordance with FED-STD-313. The safety data sheet should be submitted with each shipment of asphaltic compound as specified in FED-STD-313.

6.6 Previous Sources of Supply. The following listed manufacturers were previously approved sources. The procuring activity is cautioned that the following listed manufacturers may no longer produce a product that meets the requirements of this specification.

Government Designation	Manufacturer's designation	Test or Qualification Reference	Manufacturer's Name and Address
MIL-C-82604	"Pioneer 590L"	NEDE Dept. (Code 502) Jan 1973	Chemtura 4240 White Oak Ave. Hammond, IN 46320 Plant: Lawrenceville, IL www.chemtura.com/ 800.243.3024
MIL-C-82604	"Pioneer 590P"	NEDE Dept. (Code 502) Jan 1973	Chemtura 1000 Convery Bldg. Perth Amboy, NJ 06861 Plant: Perth Amboy, NJ www.chemtura.com/ 800.243.3024

6.7 Subject term (key word) listing.

Thermal insulation
High explosives

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.

Custodians:
Navy – OS

Preparing Activity:
Navy - OS
(Project. 8030-2009-004)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.daps.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T07:17Z
Check the source to verify that this is the current version before use.